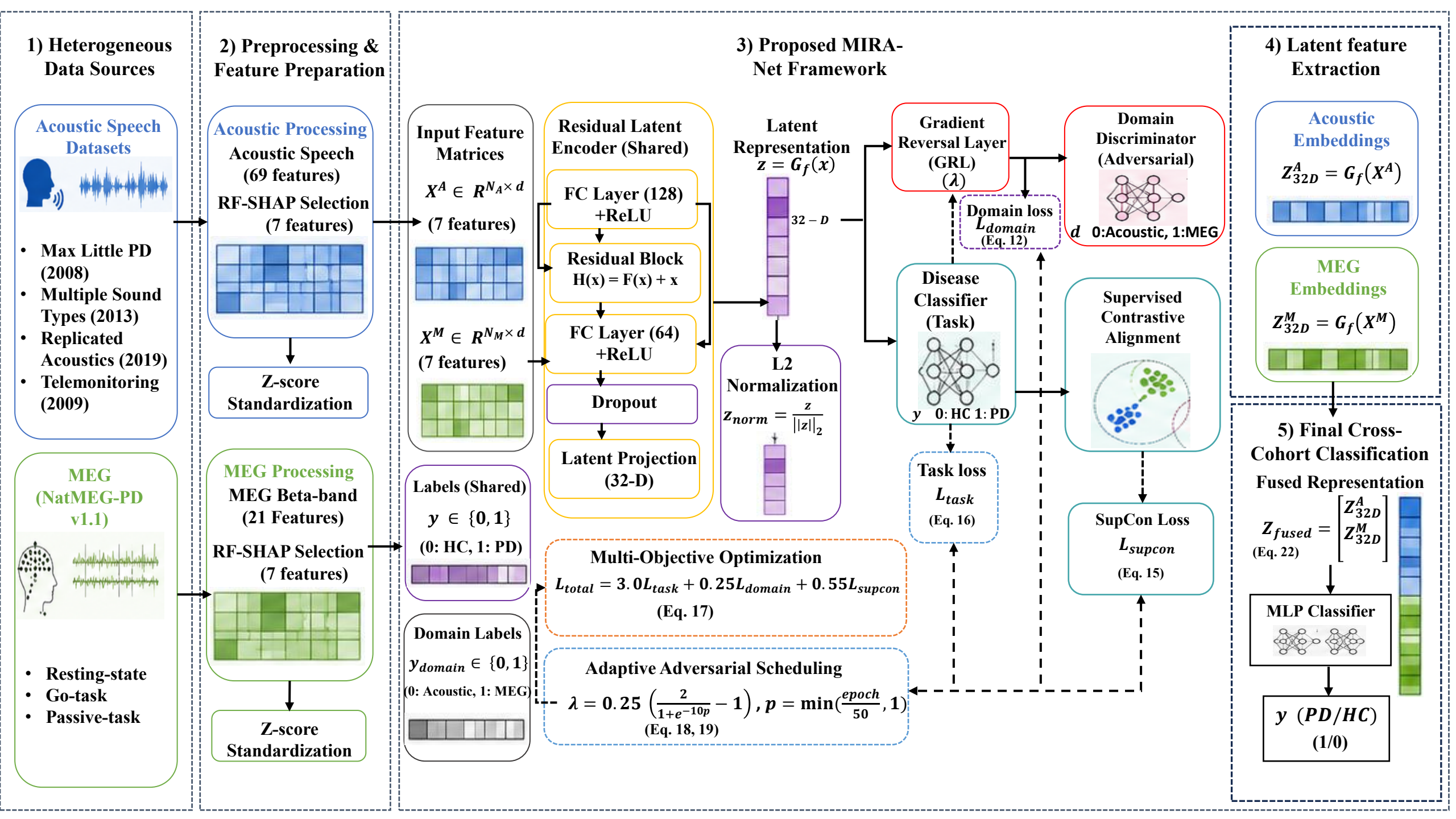

1) Heterogeneous Data Sources
Acoustic Speech Datasets
Max Little PD (2008)
Multiple Sound Types (2013)
Replicated Acoustics (2019)
Telemonitoring (2009)
MEG (NatMEG-PD v1.1)
Resting-state
Go-task
Passive-task
2) Preprocessing & Feature Preparation
Acoustic Processing
Acoustic Speech (69 features)
RF-SHAP Selection (7 features)
Z-score Standardization
MEG Processing
MEG Beta-band (21 Features)
RF-SHAP Selection (7 features)
Z-score Standardization
3) Proposed MIRA-Net Framework
Input Feature Matrices
$X^A \in R^{N_A \times d}$
(7 features)
$X^M \in R^{N_M \times d}$
(7 features)
Labels (Shared)
$y \in \{0,1\}$
(0: HC, 1: PD)
Domain Labels
$y_{domain} \in \{0,1\}$
(0: Acoustic, 1: MEG)
Residual Latent Encoder (Shared)
FC Layer (128) +ReLU
Residual Block H(x) = F(x) + x
FC Layer (64) +ReLU
Dropout
Latent Projection (32-D)
Latent Representation $z = G_f(x)$
32 – D
L2 Normalization
$z_{norm} = \frac{z}{||z||_2}$
Gradient Reversal Layer (GRL) (λ)
Domain loss L_{domain} (Eq. 12)
Domain Discriminator (Adversarial)
d 0:Acoustic, 1:MEG
Disease Classifier (Task)
y 0: HC 1: PD
Task loss L_{task} (Eq. 16)
Supervised Contrastive Alignment
SupCon Loss L_{supcon} (Eq. 15)
Multi-Objective Optimization
$L_{total} = 3.0L_{task} + 0.25L_{domain} + 0.55L_{supcon}$
(Eq. 17)
Adaptive Adversarial Scheduling
$\lambda = 0.25\left(\frac{2}{1+e^{-10p}} - 1\right), p = \min(\frac{epoch}{50}, 1)$
(Eq. 18, 19)
4) Latent feature Extraction
Acoustic Embeddings
$Z^A_{32D} = G_f(X^A)$
MEG Embeddings
$Z^M_{32D} = G_f(X^M)$
5) Final Cross-Cohort Classification
Fused Representation
$Z_{fused} = \begin{bmatrix} Z^A_{32D} \\ Z^M_{32D} \end{bmatrix}$
(Eq. 22)
MLP Classifier
y (PD/HC) (1/0)